The Multiplicity Bundle Theorem: Schur–Weyl Duality, Crystal Invisibility, and the Coboundary Hierarchy

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Abstract

We prove that the coboundary commutor on tensor powers of the fundamental representation $V = \mathbb{C}^n$ of GL_n, constructed at $q = 0$ from the swap operator, factorises through the Schur–Weyl decomposition as $\sigma_{[i,j]} = \bigoplus_\lambda (\mathrm{id}_{V_\lambda} \otimes \rho_{S_\lambda}(w_0^{[i,j]}))$. The crystal functor sends this commutor to the identity: the multiplicity spaces S_λ (Specht modules) form a "multiplicity bundle" that the crystal functor forgets. This explains why the Henriques–Kamnitzer crystal commutor is trivial on identical fundamental factors and completes the vertical axis of the categorical phase diagram: braided monoidal $\to$ coboundary monoidal $\to$ symmetric monoidal corresponds to forgetting the spectral parameter, then the multiplicity bundle.

1 Setup and Notation

Throughout, $\mathfrak{g} = \mathfrak{sl}_n$ with $n \geq 2$, $V = \mathbb{C}^n$ the fundamental (vector) representation of GL_n, and $k \geq 2$ an integer.

Definition 1 (Schur–Weyl decomposition)**.** *The tensor power $V^{\otimes k}$ carries commuting actions of GL_n (diagonal action on each factor) and S_k (permutation of tensor positions). Schur–Weyl duality gives the isotypic decomposition*

$$V^{\otimes k} \cong \bigoplus_{\substack{\lambda \vdash k \\ \ell(\lambda) \leq n}} V_\lambda \otimes S_\lambda \tag{1}$$

as a $(\mathrm{GL}_n \times S_k)$-bimodule, where V_λ is the irreducible GL_n-module of highest weight λ and S_λ is the Specht module (irreducible S_k-module) indexed by λ.

Definition 2 (Coboundary commutor at $q = 0$)**.** *For $1 \leq i < j \leq k$, the* commutor $\sigma_{[i,j]} \in \mathrm{End}(V^{\otimes k})$ *is the operator that reverses tensor factors i through j:*

$$\sigma_{[i,j]}(v_1 \otimes \cdots \otimes v_k) = v_1 \otimes \cdots \otimes v_{i-1} \otimes v_j \otimes v_{j-1} \otimes \cdots \otimes v_i \otimes v_{j+1} \otimes \cdots \otimes v_k.$$

Equivalently, $\sigma_{[i,j]}$ is the image of the longest element $w_0^{[i,j]} \in S_{\{i,\ldots,j\}} \hookrightarrow S_k$ under the permutation representation $S_k \to \mathrm{GL}(V^{\otimes k})$. By the Cactus Representation Theorem [1], the map $s_{[i,j]} \mapsto \sigma_{[i,j]}$ defines a representation of the cactus group J_k.

Definition 3 (Crystals)**.** *The crystal $B(\omega_1)$ of the fundamental representation is the set $\{1, 2, \ldots, n\}$ with Kashiwara operators $\tilde{f}_i(j) = j + 1$ if $j = i$, and undefined otherwise (and dually for $\tilde{e}_i$). The tensor product crystal $B(\omega_1)^{\otimes k}$ has elements $(b_1, \ldots, b_k)$ with $b_\ell \in \{1, \ldots, n\}$ and Kashiwara operators defined by the* signature rule.

Definition 4 (Combinatorial R-matrix)**.** *For crystals B_1, B_2 of integrable modules, the* combinatorial R-matrix $\sigma^{\mathrm{comb}}_{B_1,B_2} : B_1 \otimes B_2 \xrightarrow{\sim} B_2 \otimes B_1$ *is the unique crystal isomorphism that identifies corresponding connected components. Its existence follows from the fact that $B_1 \otimes B_2$ and $B_2 \otimes B_1$ have the same decomposition into irreducible components with the same multiplicities (see [4]). When $B_1 = B_2$, both sides are the same crystal, so σ^{comb} is a crystal automorphism of $B_1 \otimes B_1$.*

2 The Multiplicity Bundle Theorem

Theorem 5 (Multiplicity Bundle)**.** *Let $V = \mathbb{C}^n$, $k \geq 2$, and $\sigma_{[i,j]}$ the coboundary commutor of Definition 2. Then:*

(1) ***Schur–Weyl factorisation.*** *Under the decomposition* (1),

$$\sigma_{[i,j]} = \bigoplus_{\lambda} \big(\mathrm{id}_{V_\lambda} \otimes \rho_{S_\lambda}(w_0^{[i,j]})\big), \tag{2}$$

where $\rho_{S_\lambda} : S_k \to \mathrm{GL}(S_\lambda)$ is the Specht module representation.

(2) ***Crystal invisibility.*** *The crystal functor $F : \mathrm{Rep}(\mathrm{GL}_n) \to \mathrm{Crys}(\mathfrak{sl}_n)$ sends $\sigma_{[i,j]}$ to the identity:*

$$F(\sigma_{[i,j]}) = \mathrm{id}_{B(\omega_1)^{\otimes k}}.$$

(3) ***Multiplicity bundle.*** *The fibre of F over each connected component $B(\lambda) \subset B(\omega_1)^{\otimes k}$ is the Specht module S_λ. The coboundary commutor acts nontrivially only on these fibres.*

(4) ***Three-level hierarchy.*** *The sequence*

$$\mathrm{Rep}(U_q(\mathfrak{sl}_n)) \xrightarrow{q\to 0} \mathrm{Rep}(U_0(\mathfrak{sl}_n)) \xrightarrow{F} \mathrm{Crys}(\mathfrak{sl}_n)$$

corresponds to

$$\text{braided monoidal} \xrightarrow{\text{forget spectral parameter}} \text{coboundary monoidal} \xrightarrow{\text{forget multiplicity}} \text{symmetric monoidal.}$$

3 Proof

3.1 Part (1): Schur–Weyl factorisation

Proof. The commutor $\sigma_{[i,j]}$ is, by definition, the image of the permutation $w_0^{[i,j]} \in S_k$ under the representation $\rho : S_k \to \mathrm{GL}(V^{\otimes k})$ that permutes tensor positions. By Schur–Weyl duality (1), this representation decomposes as

$$\rho(w) = \bigoplus_{\lambda} \big(\mathrm{id}_{V_\lambda} \otimes \rho_{S_\lambda}(w)\big) \qquad \text{for all } w \in S_k.$$

(The GL_n-factor V_λ is inert under S_k because the two actions commute, and Schur's lemma forces the S_k-action on the V_λ-isotypic component to be $\mathrm{id}_{V_\lambda} \otimes A$ for some $A \in \mathrm{End}(S_\lambda)$; the bimodule structure identifies $A = \rho_{S_\lambda}(w)$.)

Setting $w = w_0^{[i,j]}$ gives (2). □

3.2 Part (2): Crystal invisibility

The proof proceeds in three steps: we establish a rigidity property of crystals, use it to show the combinatorial R-matrix on identical fundamental crystals is trivial, then extend to general commutors.

Lemma 6 (Crystal automorphism rigidity)**.** *Let $B(\mu)$ be a connected crystal of highest weight μ. Then* $\mathrm{Aut}_{\mathrm{Crys}}(B(\mu)) = \{\mathrm{id}\}$.

Proof. Let $\varphi\colon B(\mu) \to B(\mu)$ be a crystal automorphism. Since φ preserves weights, φ must fix the unique highest weight element u_μ (the only element of weight μ). Every other element $b \in B(\mu)$ is obtained from u_μ by a sequence of Kashiwara lowering operators: $b = \tilde{f}_{i_1} \cdots \tilde{f}_{i_r}(u_\mu)$. Since φ commutes with Kashiwara operators:

$$\varphi(b) = \tilde{f}_{i_1} \cdots \tilde{f}_{i_r}(\varphi(u_\mu)) = \tilde{f}_{i_1} \cdots \tilde{f}_{i_r}(u_\mu) = b.$$

Therefore $\varphi = \mathrm{id}$. □

Lemma 7 (Multiplicity-free tensor product)**.** *For type A_{n-1} ($n \geq 2$), the tensor product of crystals $B(\omega_1) \otimes B(\omega_1)$ is multiplicity-free:*

$$B(\omega_1) \otimes B(\omega_1) \cong B(2\omega_1) \sqcup B(\omega_2),$$

where $B(2\omega_1)$ corresponds to the symmetric square and $B(\omega_2)$ to the alternating square. (For $n = 2$, $B(\omega_2) = B(0)$ is a single point.)

Proof. This is the crystal analogue of the classical decomposition $\mathrm{Sym}^2(V) \oplus \bigwedge^2(V) = V \otimes V$ for $V = \mathbb{C}^n$. The representation $\mathrm{Sym}^2(V)$ has highest weight $2\omega_1$ and $\bigwedge^2(V)$ has highest weight ω_2 (or 0 when $n = 2$). Both appear with multiplicity one, so the crystal decomposition is multiplicity-free. □

Proposition 8 (Trivial combinatorial R-matrix on identical fundamentals)**.** *The combinatorial R-matrix $\sigma^{\mathrm{comb}}\colon B(\omega_1) \otimes B(\omega_1) \to B(\omega_1) \otimes B(\omega_1)$ is the identity map.*

Proof. Since source and target are the same crystal $B = B(\omega_1)^{\otimes 2}$, the combinatorial R-matrix is a crystal automorphism of B.

By Lemma 7, $B \cong B(2\omega_1) \sqcup B(\omega_2)$, with each irreducible appearing exactly once. Any crystal automorphism of B must preserve connected components (since it preserves weights and Kashiwara operators). With multiplicity one for each component, the automorphism restricted to each component $B(\mu)$ is a crystal automorphism of $B(\mu)$, which is the identity by Lemma 6. Therefore $\sigma^{\mathrm{comb}} = \mathrm{id}$. □

Proof of Part (2). We prove a stronger statement: the *crystal commutor* $\sigma^{\mathrm{crys}}_{[i,j]}$ on $B(\omega_1)^{\otimes k}$, defined intrinsically in the crystal category, is the identity for all $1 \leq i < j \leq k$.

Step 1: Pairwise crystal commutor. The crystal commutor on $B(\omega_1)^{\otimes k}$ is determined by the pairwise combinatorial R-matrices on adjacent factors. The commutor $\sigma^{\mathrm{crys}}_{[i,i+1]}$ acts by the combinatorial R-matrix on factors i and $i+1$ (both copies of $B(\omega_1)$), leaving other factors unchanged. By Proposition 8, $\sigma^{\mathrm{crys}}_{[i,i+1]} = \mathrm{id}$ for each i.

Step 2: General intervals. The crystal commutor $\sigma^{\mathrm{crys}}_{[i,j]}$ (reversal of positions i through j) is built from pairwise commutors: the reversal permutation $w_0^{[i,j]}$ factors as a product of adjacent

transpositions, and the cactus group action on crystals is constructed from pairwise combinatorial R-matrices. Since each pairwise commutor on identical $B(\omega_1)$ factors is the identity (Step 1), any composition of them is also the identity. Therefore $\sigma^{\mathrm{crys}}_{[i,j]} = \mathrm{id}$.

Step 3: Connection to representations. At the representation level, the coboundary commutor $\sigma_{[i,j]}$ acts nontrivially (Part (1): it acts by $\rho_{S_\lambda}(w_0^{[i,j]})$ on multiplicity spaces). The passage to crystals forgets this action: the crystal "sees" only the V_λ-component (the connected component type $B(\lambda)$), not the S_λ-component (the multiplicity label). Since $\sigma_{[i,j]}$ acts as id on each V_λ, its crystal shadow is trivial.

More precisely: the Henriques–Kamnitzer crystal commutor [2] coincides with the $q \to 0$ limit of the Kamnitzer–Tingley unitarised R-matrix commutor [3]. At $q = 0$, the unitarised commutor becomes the swap P (the cactus representation of [1]), whose crystal shadow is $\sigma^{\mathrm{crys}} = \mathrm{id}$ by Steps 1–2. □

Remark 9. The triviality of $F(\sigma)$ depends crucially on the tensor factors being *identical* copies of $B(\omega_1)$. For non-identical representations $B(\lambda) \otimes B(\mu)$ with $\lambda \neq \mu$, the combinatorial R-matrix is generally nontrivial (it is the crystal analogue of the braiding/commutor). Our theorem applies specifically to $V^{\otimes k} = V^{\otimes k}$, the setting of Schur–Weyl duality.

3.3 Part (3): Multiplicity bundle

Proof. The crystal $B(\omega_1)^{\otimes k}$ decomposes into connected components:

$$B(\omega_1)^{\otimes k} = \bigsqcup_{\substack{\lambda \vdash k \\ \ell(\lambda) \leq n}} \bigsqcup_{t=1}^{f^\lambda} B_t(\lambda),$$

where $f^\lambda = \dim S_\lambda = |\,\mathrm{SYT}(\lambda)|$ is the number of standard Young tableaux of shape λ, and each $B_t(\lambda)$ is a connected component isomorphic to $B(\lambda)$.

In the Schur–Weyl decomposition (1), the crystal basis of $V^{\otimes k}$ is a set $\{G(b) : b \in B(\omega_1)^{\otimes k}\}$ where $G(b)$ is Kashiwara's global (canonical) basis element indexed by b. For $b \in B_t(\lambda)$, the element $G(b)$ lies in the isotypic component $V_\lambda \otimes S_\lambda$. Specifically:

- The crystal position within $B(\lambda)$ (i.e., which element of $B_t(\lambda)$ we have) determines the V_λ-component.
- The copy index $t \in \{1, \ldots, f^\lambda\}$ determines the S_λ-component.

This gives a *multiplicity bundle* structure:

Base space:	$B(\omega_1)^{\otimes k}$ (the crystal, a discrete set)
Fibre over $B(\lambda)$:	S_λ (the Specht module, dim f^λ)
Structure group:	S_k acting on S_λ
Connection:	Crystal operators $\tilde{e}_i, \tilde{f}_i$ (act on base, trivially on fibre)
Commutor:	$\sigma_{[i,j]}$ acts as $\rho_{S_\lambda}(w_0^{[i,j]})$ on the fibre

By Part (1), $\sigma_{[i,j]}$ acts as id_{V_λ} on the base direction (within each $B(\lambda)$) and as $\rho_{S_\lambda}(w_0^{[i,j]})$ on the fibre S_λ. By Part (2), the crystal functor F projects to the base, discarding the fibre. Therefore $F(\sigma_{[i,j]}) = \mathrm{id}$: the commutor is invisible to crystals because it lives entirely in the fibre. □

3.4 Part (4): Three-level hierarchy

Proof. We describe the three levels and what is forgotten at each transition.

Level 1: Braided monoidal (q generic). At generic q, the quantum group $U_q(\mathfrak{sl}_n)$ provides a braiding on $\mathrm{Rep}(U_q(\mathfrak{sl}_n))$ via the universal R-matrix. For the fundamental representation, the R-matrix $R(u, q)$ depends on a spectral parameter u and deformation parameter q, and satisfies the Yang–Baxter equation. The braiding is nontrivial: $\beta_{V,V} \neq \beta_{V,V}^{-1}$ in general. This structure supports *solvable* lattice models (row-to-row transfer matrices commute).

Transition $q \to 0$: forget spectral parameter. At $q = 0$, the R-matrix degenerates to the swap P, losing its dependence on the spectral parameter. The Yang–Baxter equation degenerates: the braid group representation collapses to a symmetric group representation. The residual algebraic structure is the coboundary (cactus group) structure established in [1].

Level 2: Coboundary monoidal ($q = 0$). The commutors $\sigma_{[i,j]}$ generate a J_k-action on $V^{\otimes k}$, factoring through the Specht module representations (Part (1)). The commutor is nontrivial—it acts on the multiplicity bundle $\{S_\lambda\}$—but involutive ($\sigma^2 = \mathrm{id}$, unlike the braiding at generic q).

Transition F: forget multiplicity bundle. The crystal functor F discards the fibre S_λ, retaining only the crystal $B(\lambda)$. By Part (2), $F(\sigma_{[i,j]}) = \mathrm{id}$, so the commutor becomes trivial.

Level 3: Symmetric monoidal (crystals). The category $\mathrm{Crys}(\mathfrak{sl}_n)$ with the trivial commutor (identity on $B \otimes B$ for identical crystals) is symmetric monoidal. The tensor product of identical fundamental crystals has a trivial braiding: $\sigma_{B,B}^{\mathrm{crys}} = \mathrm{id}$.

In summary:

	Category	**Structure**	**Commutor**
q generic	$\mathrm{Rep}(U_q)$	braided	R-matrix
$q = 0$	$\mathrm{Rep}(\mathrm{GL}_n)$	coboundary	$\mathrm{id} \otimes \rho_{S_\lambda}(w_0)$
Crystal	Crys	symmetric	id
Forgotten at each step:		spectral param.	multiplicity bundle

Each row retains strictly less structure than the one above. □

4 Computational Verification

Example 10 ($\mathfrak{sl}_2$, $k = 3$). Take $V = \mathbb{C}^2$, so $V^{\otimes 3} \cong \mathbb{C}^8$. The Schur–Weyl decomposition (with $\ell(\lambda) \leq 2$) gives:

$$V^{\otimes 3} \cong V_{(3)} \otimes S_{(3)} \oplus V_{(2,1)} \otimes S_{(2,1)}.$$

Here $\dim V_{(3)} = 4$ (the spin-3/2 representation), $\dim S_{(3)} = 1$, $\dim V_{(2,1)} = 2$ (the fundamental), $\dim S_{(2,1)} = 2$.

The commutor $\sigma_{[1,3]}$ (full reversal) acts as:

- On $V_{(3)} \otimes S_{(3)}$ ($\dim 4 \times 1 = 4$): $\mathrm{id} \otimes \rho_{S_{(3)}}(w_0) = \mathrm{id} \otimes (+1) = \mathrm{id}$. Contributes 4 eigenvalues of $+1$.
- On $V_{(2,1)} \otimes S_{(2,1)}$ ($\dim 2 \times 2 = 4$): $\mathrm{id}_2 \otimes M$ where $M = \rho_{S_{(2,1)}}((1\,3))$ has eigenvalues $+1, -1$. Contributes 2 eigenvalues of $+1$ and 2 of -1.

Total eigenvalues of $\sigma_{[1,3]}$: $+1$ with multiplicity $4+2=6$, -1 with multiplicity 2. This is confirmed computationally.

The crystal $B(\omega_1)^{\otimes 3}$ decomposes as $B(3\omega_1)\sqcup B(\omega_1)\sqcup B(\omega_1)$ (one copy of $B(3\omega_1)$ with 4 elements, two copies of $B(\omega_1)$ with 2 elements each, total $4+2+2=8$). The multiplicity of $B(\omega_1)$ is $2 = f^{(2,1)} = |\operatorname{SYT}(2,1)|$, matching the Specht module dimension.

Example 11 ($\mathfrak{sl}_3$, $k=3$). Take $V=\mathbb{C}^3$, so $V^{\otimes 3}\cong\mathbb{C}^{27}$. Partitions of 3 with $\ell(\lambda)\le 3$: (3), $(2,1)$, $(1,1,1)$. The decomposition:

$$V^{\otimes 3}\cong V_{(3)}\otimes S_{(3)}\oplus V_{(2,1)}\otimes S_{(2,1)}\oplus V_{(1,1,1)}\otimes S_{(1,1,1)}.$$

Dimensions: $\dim V_{(3)}=10$, $\dim V_{(2,1)}=8$, $\dim V_{(1,1,1)}=1$. Specht dimensions: $f^{(3)}=1$, $f^{(2,1)}=2$, $f^{(1,1,1)}=1$. Total: $10\cdot 1+8\cdot 2+1\cdot 1=27$. ✓

The crystal $B(\omega_1)^{\otimes 3}$ has $3^3=27$ elements, decomposing as $B(3\omega_1)\sqcup B(\omega_1+\omega_2)^{\sqcup 2}\sqcup B(0)$ (using that $V_{(2,1)}$ for GL_3 has highest weight $\omega_1+\omega_2$ and $V_{(1,1,1)}\cong\det$ has highest weight $\omega_1+\omega_2+\omega_3=0$ as an $\mathfrak{sl}_3$-weight). Crystal multiplicities: $1,2,1$, matching Specht dimensions.

5 Discussion

Remark 12 (What each level sees). The three levels of the hierarchy can be understood through the lens of the Schur–Weyl decomposition $V^{\otimes k}\cong\bigoplus_\lambda V_\lambda\otimes S_\lambda$:

- **Braided** (generic q): sees the full tensor product $V_\lambda\otimes S_\lambda$, with the R-matrix mixing both factors.
- **Coboundary** ($q=0$): sees both factors, but the commutor acts only on S_λ (Part (1)).
- **Symmetric** (crystal): sees only $B(\lambda)$, the shadow of V_λ. The multiplicity S_λ is invisible (Part (2)).

Remark 13 (Relation to Henriques–Kamnitzer). Henriques and Kamnitzer [2] defined the crystal commutor using the Schützenberger involution and showed it gives a coboundary structure on $\operatorname{Crys}(\mathfrak{g})$. For identical fundamental factors, our theorem explains *why* their commutor is trivial: it is the crystal shadow of the Schur–Weyl commutor, which lives entirely in the multiplicity bundle that the crystal functor forgets.

Remark 14 (Beyond the fundamental representation). Parts (1)–(2) extend to tensor products of distinct representations, but the conclusion changes. For $V_\lambda\otimes V_\mu$ with $\lambda\ne\mu$, the combinatorial R-matrix is generally nontrivial—the crystal commutor permutes the factors in a nontrivial way (it is the crystal analogue of the Weyl group action). The multiplicity-freeness of $B(\omega_1)^{\otimes 2}$ (Lemma 7) is special to the fundamental representation.

References

References

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